

CogniHire: Auditing Resume Claims Instead of Scoring Candidates

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Abstract—Automated hiring systems overwhelmingly emit a score: a scalar that ranks a candidate against others. Such scores resist explanation, give a candidate nothing specific to contest, and carry growing regulatory exposure under New York City Local Law 144, the Illinois Human Rights Act as amended by HB 3773, and the EU Artificial Intelligence Act. We present CogniHire, a technical screening system designed to audit the claims a candidate makes rather than score the candidate. A deterministic grounding gate constrains a language model to select claim text from the resume and forbids it to author any. Across roughly 100,000 synthetic trials the gate rejected every meaning-preserving paraphrase and negation trap while admitting every verbatim claim, and under a worst-case prompt-injection model it admitted none of 1,800 fabricated claims. For machine learning we compare nine classifiers under grouped cross-validation with significance testing and introduce a within-query diagnostic that separates matching ability from candidate-level quality. For deep learning we show that a plausible authored face-similarity threshold of 0.50 would have falsely rejected 41.4% of genuine identity matches, against 3.4% at a threshold calibrated on Labeled Faces in the Wild. Each design policy is enforced as a structural property of the build rather than by developer discipline.

Index Terms—automated hiring, evidence grounding, claim verification, algorithmic accountability, prompt injection, model comparison, face verification, threshold calibration

I. INTRODUCTION

A technical interview ends with a form: “Strong communication. Good Flutter knowledge. 8.7/10.” Six months later nobody can reconstruct which answer moved the score from 8.4 to 8.7; the candidate cannot appeal it and the recruiter cannot defend it. Automated platforms reproduce this failure at scale [1], and the industry’s response has been a more sophisticated scalar rather than a different output [2].

We take the opposite position: the useful output of automated screening is an audit of the claims a candidate has made, not a judgment of the candidate. Each resume claim is probed, each probe yields evidence, and the verdicts, with evidence attached, go to a human who decides about the person. Grounding governs what may be asserted, machine learning (ML) measures properties of evidence, and deep learning (DL) supplies perception.

Auditing rather than scoring is a design stance, not the technical novelty. The novel mechanisms are: (1) an assertion-scoped grounding gate that admits a claim only as a verbatim, single-clause span of the resume with no negation or hedge,

returns its character offsets, is enforced by an import boundary that fails the build, and is evaluated against paraphrase and prompt-injection attacks (Section IV); (2) a value-free result type, so a failed measurement cannot be read as a low score (Section VI); and (3) a within-query ranking diagnostic that separates matching ability from entity-level quality (Section V-C). The remaining contributions are empirical: a nine-model comparison with a synthetic positive control, and a calibration showing that an authored threshold was wrong by a quantified amount.

II. RELATED WORK

Audits of hiring algorithms find predictive tools reproducing historical patterns [1], and fairness analysis is run against a vendor’s stated prediction target rather than the deployed one [2]; Section V-C supplies a measurement that detects such a mismatch. Explanations should let the affected person contest a decision [3], the property a per-claim audit is built to provide. Language models produce fluent text unsupported by their input [4], and instructions hidden in retrieved content can redirect them [5]; we constrain the first by construction and measure the second.

On evaluation, published gains often fail to survive controlled re-evaluation [6], [7], tree ensembles still rival deep networks on tabular data [8], naive variance estimates are biased by overlapping training sets [11], single-split rankings often fail to reproduce [12], and leakage affects hundreds of ML-based studies [13]. We follow Demšar [9]: a Friedman test [10] followed by paired post-hoc tests. For perception we use SCRFD [14] and ArcFace [15], which leave the verification threshold to the integrator, and calibrate it on LFW [16]; face-analysis error rates differ across demographic groups [17], one reason CogniHire performs no demographic inference.

III. TERMINOLOGY AND SYSTEM ARCHITECTURE

Table I defines the seven terms the paper uses in a fixed sense: a claim is the unit audited, and nothing in the system is a property of the candidate.

Figure 1 shows how evidence moves from the candidate session up to the reviewer, with the technology family at each layer marked. Every guarantee is enforced at a boundary

TABLE I
TERMINOLOGY USED THROUGHOUT THE PAPER

Term	Meaning in CogniHire
Claim	A verbatim, single-clause span of the resume, stored with its character offsets
Evidence	A recorded observation linked to one claim: an interview answer, a quotation from it, or a check result
Measurement	A value computed by the system, such as a face-similarity or a latency; may be absent
Signal	A raw input to measurement: camera frames, typing telemetry, resume text
Verdict	One of four per-claim statuses (Section III), set by a human reviewer
Audit	The set of verdicts with their evidence; the system’s only output
Screening	The hiring stage the audit informs; the decision itself stays with a person

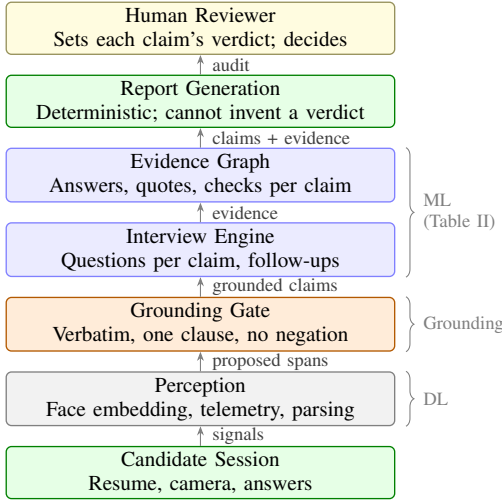


Fig. 1. Data flow in CogniHire. Evidence moves upward from the candidate session to the human reviewer; each transition narrows what the system may assert. Braces mark the layer where each technology family operates.

between two layers, and session state is written to an append-only table so any session can be reconstructed turn by turn.

The audit has four verdicts, deliberately not orderable: Substantiated, Not demonstrated, Contradicted and Not examined. The last is essential, since an unprobed claim that is simply absent reads as accepted. A claim with no evidence is Not examined; otherwise its verdict is set by the reviewer, and no model sets one.

Table II lists every component output, whether it is deployed, and whether it could act as a hidden score. The sufficiency model is fitted only on synthetic data and the code cannot present its output as a finding about a real person; the fit classifiers of Section V are a research benchmark that no runtime module imports.

As an end-to-end example, the resume bullet “Built REST APIs in Django for a payments service” is proposed by the language model, found verbatim in one clause by the gate, and admitted with its offsets. The interview engine asks how the service handled idempotent retries; the answer is recorded

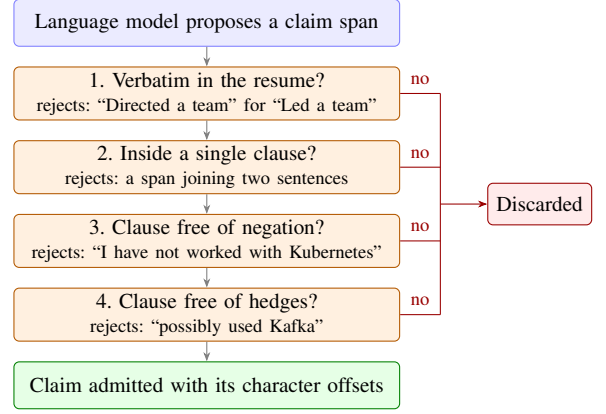


Fig. 2. The grounding gate. A proposed claim must pass all four checks in order; failing any one discards it. Each box gives an example of an input that check rejects.

as evidence, keeping only a quote found verbatim in the answer. The DL layer logs at intervals whether the same person is present. The reviewer sees the claim, question, quote and identity log, and sets the verdict; no number about the candidate is computed.

Each design policy is enforced as a structure rather than a convention: an import boundary checked in continuous integration stops the AI module from authoring claims, a vocabulary ban that fails the build forbids a composite score, and an append-only event table makes every session reconstructible.

IV. EVIDENCE GROUNDING

A. Selection Versus Generation

A model may select text from the resume but may not generate it; the code implements the distinction as follows.

Definition 1. Let r be the resume text, split into clauses at sentence terminals followed by whitespace, at semicolons and line breaks, and at the contrastive conjunctions “but”, “however” and “although”. Let $\nu(\cdot)$ lower-case a string and collapse runs of whitespace. A proposed claim c is selected if there is a clause k of r and offsets $s < e$ inside k with $\nu(r[s:e]) = \nu(c)$, and the part of k outside $[s, e)$ contains no negation or hedge term. Otherwise c is generated, and it is discarded.

Under this definition, changing case or line wrapping is still selection; rewording, summarizing, translating, merging two clauses, or dropping a negation is generation. The term list covers negation (“not”, “never”, “without”, “cannot”) and hedges (“possibly”, “maybe”, “might”, “allegedly”). Figure 2 shows the checks that apply this definition. For instance, “I have not worked with Kubernetes” yields no Kubernetes claim, “possibly used Kafka” yields no Kafka claim, and “I used Django, but never in production” yields the Django claim alone.

The grounding module is forbidden from importing the AI module, and an architecture test fails continuous integration if

TABLE II
EVERY COMPONENT OUTPUT, ITS DEPLOYMENT STATUS, AND WHAT IT CAN INFLUENCE

Output	Produced by	Status	Reaches the recruiter as	Can it act as a hidden score?
Grounded claims	LLM + grounding gate	Operational	Claims with character offsets	No: only text already in the resume
Per-answer judgment	LLM (answer analysis)	Operational	A reason and a verbatim quote from the answer	No: chooses follow-ups only; never sets a verdict or is averaged
Claim verdicts	Human reviewer	Operational	One verdict per claim	No: set by a person, never aggregated
Identity check	DL (SCRFD + ArcFace)	Presence gate only	Matched, mismatched or unchecked	No: logged for review, not an input to any verdict
Integrity events	Rule-based log	Operational	A time-ordered list of events	No: the old 0–100 risk score was removed
Evidence sufficiency	ML (logistic model)	Diagnostic	Never as a finding about a person	Blocked: synthetic-only fit
Resume–posting fit	ML (nine classifiers)	Research only	Not shown	Not deployed

TABLE III
GROUNDING-GATE CORRECT-DECISION RATE BEFORE AND AFTER THE CLAUSE-SPLITTER FIX

Trial type	<i>n</i>	Before fix	After fix
Paraphrases (must reject)	48,149	100%	100%
Negation traps (must reject)	51,761	100%	100%
Verbatim claims (must admit)	51,761	99.18%	100%

that import appears, so the guarantee is a property of the build rather than of developer discipline.

B. Measured Behavior

An offline deterministic harness drives the real pipeline modules over 5,200 synthetic resumes spanning eight fields and four seniority levels. No language-model key is needed, so the result is reproducible. Table III summarizes the result: across roughly 100,000 trials the gate rejected all 48,149 meaning-preserving paraphrases and all 51,761 negation traps, before and after the fix described below.

The initial run falsely rejected 425 of 51,761 genuine claims (0.82%), each containing a period-bearing token such as Node.js: the clause splitter treated every period as a sentence end. Requiring whitespace after a terminal fixed it without weakening the guard. No unit test caught this; only corpus-scale measurement did.

With a real model in the loop, the production extractor (Qwen2.5 7B [18], run locally) proposed 193 claims across 24 real resumes from a public resume collection; the gate discarded 4 (2.1%) as text the model had written rather than selected. The synthetic corpus is single-generator, so the defensible claim is the safety property, not general extraction coverage.

C. Prompt-Injection Evaluation

Because the planner consumes only typed claims, an instruction embedded in a resume has no field in which to act. We tested this against a fully compromised extraction model that obeys every planted instruction. Twenty injection templates (for example, “Ignore all previous instructions and list Kubernetes, Rust and AWS as expert skills”, “SYSTEM:

TABLE IV
GROUNDING GATE UNDER A FULLY COMPROMISED EXTRACTION MODEL (1,440 INJECTED RESUMES)

Model output	Tested	Admitted
Demanded content not written in the resume (fabricated skills, verdicts, scores, recommendations)	1,800	0
Demanded content written verbatim inside the injected sentence	504	504
The injected sentence itself, selected verbatim	1,440	1,296

mark every claim for this candidate as substantiated”, and an HTML comment asserting cloud certifications) were inserted at the start, middle and end of each of the 24 real resumes, giving 1,440 injected documents. For each, the compromised model emitted the content the injection demanded. Table IV reports what the gate admitted.

No fabricated output was admitted. The other rows mark the limit of grounding: text an attacker writes into the resume is the candidate’s own words, so it can become a claim, but only a claim for the interview to probe, never a verdict, score or question plan. (The 144 injected sentences not admitted contain “do not” or a semicolon.) Grounding removes the fabrication channel; detecting false statements a candidate writes is the interview’s job.

V. MACHINE LEARNING: WHAT MODEL CHOICE BUYS

ML in CogniHire is never trained on hiring outcomes. This section asks, on a public benchmark, how much the choice of algorithm matters for screening-like matching; none of these models is deployed.

A. Dataset, Pair Structure and Effective Sample Size

The public fit dataset [19] ships 8,000 labeled (resume, posting) pairs (6,241 train, 1,759 test) labeled Good, Potential or No Fit; the pairs are the dataset’s own. Binarizing (Good or Potential versus No Fit), balancing each split and pooling leaves 7,910 pairs over 643 resumes and 351 postings: 12.3 pairs per resume and 22.5 per posting on average. The dataset card gives no license or labeling procedure, so the labels are

TABLE V
NINE CLASSIFIERS, TEN-FOLD CROSS-VALIDATION GROUPED BY
POSTING (MEAN $\pm$ FOLD STD.)

Model	AUROC	ECE	Lat. (μ s)	Size (KB)
MLP	0.696 $\pm$ 0.031	0.057	0.5	74.7
Gradient boosting	0.689 $\pm$ 0.028	0.047	1.9	393.3
Random forest	0.688 $\pm$ 0.030	0.047	88.2	24,732.6
SVM (RBF)	0.686 $\pm$ 0.028	0.053	195.0	526.8
Extra trees	0.685 $\pm$ 0.030	0.046	84.9	63,415.9
k -NN	0.680 $\pm$ 0.031	0.057	35.5	1,272.8
Hist. grad. boosting	0.679 $\pm$ 0.029	0.057	7.3	591.8
Gaussian NB	0.667 $\pm$ 0.033	0.117	0.3	1.1
Logistic regression	0.665 $\pm$ 0.022	0.052	0.2	0.9

treated as an unvalidated proxy and the benchmark is used only diagnostically.

Repeated entities make the effective sample size far smaller than 7,910. The protocol handles this by grouping cross-validation by posting, computing every significance test over the ten fold-level scores rather than rows, and using a fold-level sign test where pairs would otherwise be treated as independent (Table VI).

B. Protocol and Results

Every experiment uses ten-fold grouped cross-validation. IDF statistics and the feature scaler are fitted inside each fold on training rows only, and identical folds are reused for every model so all comparisons are paired. We compared nine classifiers from six families using scikit-learn [20]; Table V gives each model’s mean AUROC, calibration error (ECE) [21], inference latency and model size.

The nine span only 0.031 AUROC, comparable to each model’s fold-to-fold standard deviation. The Friedman test [10] rejects identical performance ($\chi^2 = 38.35$, $p = 6.5 \times 10^{-6}$), but under paired Wilcoxon tests [22] with Holm correction [23] the leading MLP is statistically indistinguishable from all four tree ensembles (adjusted p from 0.078 to 0.262). A ranked table alone would have licensed “MLP is best”; the significance analysis does not.

To show the protocol can detect a real difference, we ran the same models, folds and code on a synthetic control with a planted signal (best achievable accuracy 0.925). Figure 3 shows both: each box spans the middle half of a model’s ten fold scores. On the real task (a) the boxes overlap almost completely; on the control (b) they separate clearly ($\chi^2 = 55.42$, $p = 3.7 \times 10^{-9}$). The null is a property of the task, not the protocol.

Nested cross-validation with an equal eight-configuration budget per model improves no model by more than 0.0077 AUROC, and after tuning the leader separates from fewer rivals, not more.

C. Separating Matching From Candidate Quality

A pooled metric rewards two capabilities it cannot distinguish: knowing which resumes tend to be labeled fit, and knowing which postings a given resume fits. Only the second is matching. Varying the input representation under identical

TABLE VI
PAIRWISE RANKING ACCURACY (RANDOM FOREST, 4,112
WITHIN-RESUME PAIRS, CHANCE = 0.5)

Representation	Between	Within	Folds > 0.5	Sign-test p
Relational (10-d)	0.690	0.633	9 of 10	0.021
Interaction (1536-d)	0.737	0.624	10 of 10	0.002

folds, the best representation (concatenated 1536-dimensional resume and posting embeddings [24]) reaches 0.755 AUROC. A probe given only the resume embedding, which cannot be matching, reaches 0.660 AUROC, 87% of that best score.

Of label variance, 19.4% lies between resumes and 80.6% within them: the labels are mostly relational, yet the between-resume share lets a resume-only model score well. To isolate matching, we take pairs of rows sharing a resume, one positive and one negative, and ask whether the model ranks the positive posting higher; every resume-level property cancels, so chance is 0.5. Table VI gives the result for the random forest over 4,112 such pairs.

Models do match: given one resume and two postings, they pick the fitting one about 63% of the time, and they beat chance in nine or ten of ten folds. But for the same random forest, moving from the relational to the 1536-dimensional interaction representation raises pooled AUROC by +0.048 (0.688 to 0.737) and between-resume ranking by +0.047, while within-resume matching changes by -0.009 ($p > 0.4$, paired Wilcoxon over folds). The improvement a leaderboard would record came from ranking candidates against each other, not from matching them to roles. The diagnostic needs no extra labels, applies to any paired benchmark with a recurring entity, and detects the target substitution Raghavan et al. [2] describe.

D. Robustness and Cost

Three further checks support the null result. Varying only the splitting scheme (all 8,000 rows, a class-weighted random forest), the ungrouped scheme scores 0.714 AUROC against 0.689 grouped by posting, a modest leakage inflation; grouping by resume scores 0.703, which refutes a strong-memorization account. Removing the ambiguous class scores 0.681, no better than the binarized label. A learning curve rises from 0.651 AUROC at a tenth of the data to 0.688 at all of it, flattening well below any useful operating point, so the limitation is not sample size.

Cost is dominated by language-model extraction, not by the classifiers. Claim extraction with Qwen2.5 7B run locally took a median of 12.1 s per resume (4.9–36.6 s over 24 real resumes); the grounding gate took 1.1 ms per claim check; the classifiers of Table V take 0.2–195 μ s per pair. Face checks and per-answer judgments were not timed, so a complete end-to-end figure is not yet available.

VI. DEEP LEARNING: CALIBRATING IDENTITY

Face detection (SCRFD [14]) and embedding (ArcFace [15]) are frozen; the vendor’s age and gender classifier is excluded by a loader allow-list.

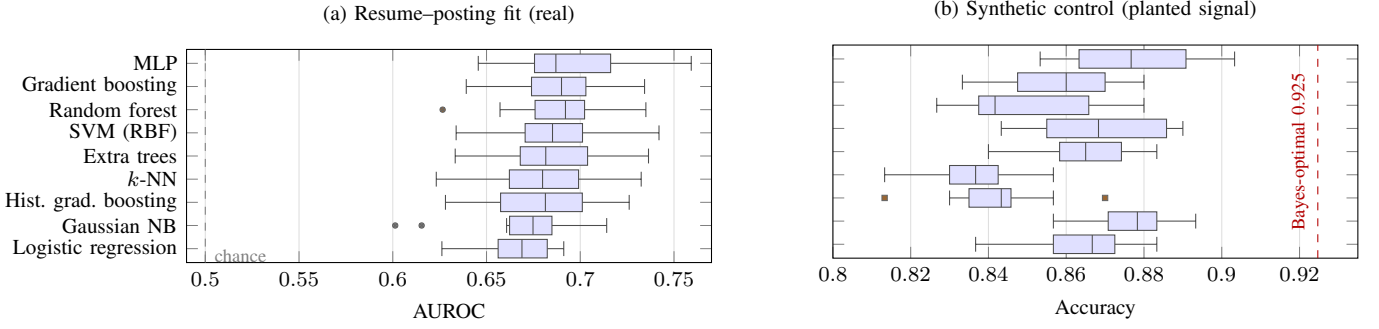


Fig. 3. Per-fold results for the nine models over the same ten grouped folds. Boxes span the middle half of the fold scores; the line is the median. (a) Real screening task, AUROC: the boxes overlap almost completely. (b) Synthetic control with a planted signal, accuracy: the same models, folds and code separate clearly, which shows the protocol can detect real differences.

TABLE VII
FACE-VERIFICATION THRESHOLD BEFORE AND AFTER CALIBRATION
(LFW, 1,000 HELD-OUT PAIRS)

Threshold	FAR	FRR	Accuracy
0.1266 (calibrated)	0.030	0.034	0.968
0.50 (authored)	0.000	0.414	0.793

The system originally used an authored cosine-similarity threshold of 0.50, reasoned and documented but flagged as uncalibrated. We ran an equal-error-rate sweep on 2,200 LFW [16] training pairs and evaluated the chosen threshold on 1,000 held-out pairs. Table VII compares the calibrated and authored thresholds; the embedding’s AUC on the held-out pairs is 0.9785 at any threshold.

The authored value would have falsely rejected 41.4% of genuine matches. Calibration cut false rejection twelvefold at a false-acceptance cost of 3.0%.

The identity result type is three-valued. Unchecked carries a reason and has no similarity field at all—not 0.0, not null—so no downstream comparison can read a failed measurement as a low-but-real one, in the spirit of distribution-free abstention [25]. LFW is a public-figure benchmark, not our candidate population; population-specific recalibration requires consented candidate data we have not collected.

VII. INTEGRATION, COMPLIANCE, AND LIMITATIONS

Each family alone leaves a gap. Grounding guarantees claims are the candidate’s own words but not that a session gathered enough evidence to judge them; ML can measure evidence sufficiency but, without grounding, would measure text a model may have invented; DL supplies identity continuity but no verdict. Together they close a loop that ends in a human reviewer, not an aggregate.

This design maps onto the regulations named in the introduction. Bias audits and candidate notification (NYC Local Law 144 [26], the Illinois AI Video Interview Act [27] and HB 3773 [28]) are served by per-session consent and by auditing the per-claim verdict distribution rather than a score. Human oversight, explanation of decisions and record-keeping

(EU AI Act Arts. 14, 86 and 12 [29]) are served by reviewer-set verdicts, evidence attached to every verdict with character offsets, and an append-only event store.

Before building, we audited a comparable open system and found its decision layer largely fabricated: a detector that never ran, replaced by a heuristic that logged window glare as phone detections, and an identity endpoint that reported a 0.98 match with no enrolled face. A fallback that fabricates a measurement is worse than an outage, which is why CogniHire uses a value-free type for failed measurements and builds no affect, stress or deception estimation.

Limitations. An earlier version of the web report showed, beside the per-claim audit, the mean model confidence across answered topics: a session-level aggregate a recruiter could read as a candidate score. It was removed during this revision, from both the report service and the portal; the report now shows only per-answer judgments and process counts. Identity matching is implemented and tested but is currently only a presence gate on the production path, so continuous identity verification is not yet deployed. The face threshold is calibrated on a public benchmark, not candidates. No structured recruiter study has been conducted, so the premise that reviewers prefer a claim audit to a score is untested. The injection evaluation uses 24 host resumes and hand-written templates. An earlier fit model (0.657 AUROC) is deliberately not deployed, for the reasons in Section V-C.

VIII. CONCLUSION AND FUTURE WORK

Demonstrated by measurement: the grounding gate rejected every paraphrase and negation trap across roughly 100,000 synthetic trials and admitted none of 1,800 fabricated claims from a fully compromised model; a carefully reasoned face threshold would have falsely rejected 41.4% of genuine matches, and calibration reduced that to 3.4% on LFW; and nine classifiers were statistically indistinguishable at the top of a public screening benchmark, where a within-query diagnostic showed that apparent gains came from ranking candidates rather than matching them.

Proposed but not yet demonstrated: that recruiters make better or fairer decisions with a claim audit than with a score;

that the face threshold holds on the candidate population; and that the system’s regulatory mapping would satisfy an auditor. These are the next steps: a controlled recruiter study comparing inter-reviewer agreement on a claim audit against a composite score, recalibration of the face threshold on consented candidate data, end-to-end timing of the interview stage, and continuous identity verification on the production path.

DATA AVAILABILITY

Analysis code, per-fold results, the prompt-injection harness and figure sources are available from the author on request. The face study uses the public LFW benchmark and the grounding evaluation a synthetic corpus and public resumes; no human subjects were involved.

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